Supplementary materials on "A class of new symmetric distributions based on scale mixtures of normal distribution and mean regression models by using N-EM and US algorithms"

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This supplementary material presents proofs of Theorems 1–3 and calculations of gradient and Hessian matrix of the $\log \text{Ge-N}(x|\boldsymbol{\theta})$ in S1; statistical inference methods for three specific scale mixtures of normal distribution in S2; real data analysis for the second real data (i.e., liver disease data) in S3; some technical derivations in S4; and a brief introduction of generalized inverse Gaussian distribution in S5.

S1. Proofs of Theorems 1–3 and calculations of gradient and Hessian matrix of the $\log$ Ge-N$(x|\boldsymbol{\theta})$

S1.1 Proofs of Theorems 1–3

Proof of Theorem 1. For consistency, we need to vertify Assumptions (A1)–(A5). Since the Ge-N pdf explicitly adopts a scale–mixture structure

$$\text{Ge-N}(x|\boldsymbol{\theta}) = \frac{1}{\sqrt{2\pi}\sigma}\int_0^\infty f_\tau(\tau|\boldsymbol{\lambda}) \cdot \sqrt{\tau}\exp\left[-\frac{\tau(x-\mu)^2}{2\sigma^2}\right] \mathrm{d}\tau, \quad x \in \mathbb{R} \tag{S1.1}$$

and $\exp[-\tau(x-\mu)^2/(2\sigma^2)]$ is an elementary function, the analytic properties of Ge-N distribution are inherited from the kernel $f_\tau(\tau|\boldsymbol{\lambda})$ inside the integral. First, Assumption (A1) merely states that the data are independent and identically distributed from the Ge-N distribution that can obviously serve (A1) by construction. Note that the support $\mathbb{S}$ of $X \sim$ Ge-N($\boldsymbol{\theta}$) is $\mathbb{R}$, so Assumption (A2) is automatically satisfied.

Regarding Assumption (A3), provided that the mixing kernel $f_\tau(\tau|\boldsymbol{\lambda})$ generates mutually singular probability measures for distinct hyper-parameters $\boldsymbol{\lambda}$, the integrands differ on a measurable set whenever $\boldsymbol{\theta}_1 \neq \boldsymbol{\theta}_2$. Integration over the measurable set then produces marginal densities Ge-N($x|\boldsymbol{\theta}_1$) and Ge-N($x|\boldsymbol{\theta}_2$) which are different on a Borel set with positive Lebesgue measure, hence the mapping $\boldsymbol{\theta} \mapsto$ Ge-N($\dot{|}\boldsymbol{\theta}$) is injective, guaranteeing global identifiability and, consequently, the validity of (A3). The parameter space is $\boldsymbol{\Theta} \triangleq \mathbb{R} \times \mathbb{R}_+ \times \boldsymbol{\Lambda}$, which is manifestly an open space to satisfy (A4).

Regarding Assumption (A5), the exponential term $\sqrt{\tau}\exp[-\tau(x-\mu)^2/(2\sigma^2)]$ is infinitely smooth in μ and σ^2 with bounded derivatives on compact τ intervals. If $f_\tau(\tau|\boldsymbol{\lambda})$ is once differentiable in $\boldsymbol{\lambda}$ with integrable derivatives, we can differentiate under the integral sign, giving a continuously differentiable likelihood and securing (A5). Then we have $\hat{\boldsymbol{\theta}}_n \xrightarrow{\mathrm{P}} \boldsymbol{\theta}_0$. $\square$

Proof of Theorem 2. As for asymptotic normality, the third–order differentiability and continuity of the Ge-N density in $\boldsymbol{\theta}$ hinge on the smoothness of the mixing density $f_\tau(\tau|\boldsymbol{\lambda})$. In fact, if $f_\tau(\tau|\boldsymbol{\lambda})$ is three times differentiable in $\boldsymbol{\theta}$ and each derivative is integrable with respect to τ, then the density Ge-N($x|\boldsymbol{\theta}$) is three times differentiable with respect to $\boldsymbol{\theta}$ and the third derivative is continuous. Moreover the integration and differentiation can be interchanged which means Assumption (A6) holds. Since Ge-N($x|\boldsymbol{\theta}$) is a smooth function of $\boldsymbol{\theta}$ (according to conditions (A5) and (A6)), its derivatives are bounded in the neighborhood of $\boldsymbol{\theta}_0$. Therefore, there exists a constant C such that:

$$\left\|\frac{\partial \log \text{Ge-N}(x|\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right\| \leqslant C, \quad \boldsymbol{\theta} \text{ in the neighborhood of } \boldsymbol{\theta}_0.$$

By the mean value theorem (Rudin 1976 [5]), if $\boldsymbol{\theta}_1$ and $\boldsymbol{\theta}_2$ are in the neighborhood of $\boldsymbol{\theta}_0$, there exists a $\boldsymbol{\theta}^*$ in the neighborhood of $\boldsymbol{\theta}_0$ such that

$$\begin{aligned}
|\log \text{Ge-N}(x|\boldsymbol{\theta}_1) - \log \text{Ge-N}(x|\boldsymbol{\theta}_2)| &= \left|\frac{\partial \log \text{Ge-N}(x|\boldsymbol{\theta})}{\partial \boldsymbol{\theta}^\top}\Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}^*} \cdot (\boldsymbol{\theta}_2 - \boldsymbol{\theta}_1)\right| \\
&\leqslant \left\|\frac{\partial \log \text{Ge-N}(x|\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\Big|_{\boldsymbol{\theta}=\boldsymbol{\theta}^*}\right\| \cdot \|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\| \leqslant C\,\|\boldsymbol{\theta}_1 - \boldsymbol{\theta}_2\|\,.
\end{aligned}$$

Since $M(x) = C$ is a constant, $E[M^2(X)] = C^2 < \infty$ which satisfies Lipschitz condition. Regarding the non-singularity of the information matrix, Ge-N$(x|\boldsymbol{\theta})$ is a smooth function of $\boldsymbol{\theta}$ and its derivatives are bounded in the neighborhood of $\boldsymbol{\theta}_0$, we have $\mathbf{J}(\boldsymbol{\theta}_0)$ is finite. Moreover, the model is identifiable (Assumption (A3)), different parameters correspond to different distributions, and thus $\mathbf{J}(\boldsymbol{\theta}_0)$ is nonsingular. Finally, we have the MLE $\hat{\boldsymbol{\theta}}_n = \hat{\boldsymbol{\theta}}_n(X_1, \ldots, X_n)$ is asymptotically normal, i.e., $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{\text{D}} N_{2+m}(\mathbf{0}, \mathbf{J}^{-1}(\boldsymbol{\theta}_0))$ as $n \to \infty$. □

Proof of Theorem 3. We prove Theorem 3 based on the scale mixture of normal representation and standard integral asymptotic techniques (i.e., the Laplace method and Watson's lemma). Three distributions belong to the symmetric generalized scale-mixture normal family with density Ge-N$(x|\boldsymbol{\theta})$ given by (S1.1), where τ denotes the mixing variable. Without loss of generality, we set $\mu = 0$ and $\sigma = 1$.

(a) For the IGa-N distribution, the mixing variable $\tau \sim \text{Gamma}(\lambda, 1)$. Substituting the gamma density into (S1.1) and applying the Laplace steepest-descent method for large $|x|$, the dominant exponential term governs the tail decay, while the polynomial term provides minor-order correction. Standard asymptotic expansion yields

$$f_X(x) \asymp \frac{2^{-\lambda/2}}{\sigma\,\Gamma(\lambda)} \left(\frac{|x-\mu|}{\sigma}\right)^{\lambda-1} \exp\left(-\frac{\sqrt{2}\,|x-\mu|}{\sigma}\right) \quad \text{as} \quad |x| \to \infty.$$

Since the tail decays exponentially, all absolute moments of the IGa-N distribution are finite. Direct calculation gives $\text{Var}(X) = \sigma^2\lambda$ and kurtosis $\text{Kurt}(X) = 3/\lambda$.

(b) The IGau-N model is a semi-heavy exponential-tailed scale-mixture normal distribution, whose mixing variable follows an inverse Gaussian distribution parameterized by λ. Different from the standard normal–inverse Gaussian distribution — a standard two-parameter exponential-tailed scale-mixture normal widely used in financial and actuarial modeling with tail prefactor $|x|^{-3/2}$ — the IGau-N model possesses a slower-decaying polynomial prefactor $|x|^{-1/2}$. By applying the Laplace asymptotic method to the mixing integral, the tail asymptotic is dominated by the exponential tempering term $\exp(-|x-\mu|/\sigma)$, means that the exponential decay term asymptotically suppresses extreme tail probabilities, which eliminates the divergent moment behavior of polynomial-tailed distributions. The exact tail asymptotic form is

$$f_Y(x) \asymp \frac{e^{1/\lambda}}{\sqrt{2\pi\sigma}} |x-\mu|^{-\frac{1}{2}} \exp\left(-\frac{|x-\mu|}{\sigma}\right) \quad \text{as} \quad |x| \to \infty.$$

Owing to the exponential tail truncation, all absolute moments of the IGau-N distribution exist finitely. Straightforward computation yields $\mathrm{Var}(Y) = \sigma^2(1+1/\lambda)$.

(c) For the IBeta-N model, the mixing variable follows a beta-prime distribution with parameters (λ_1, λ_2). We apply Watson's lemma for polynomial integral asymptotically, which characterizes power-law tail behaviors for scale-mixture normal models. The mixing density decays polynomially near zero, leading to polynomial tail decay of the resulting pdf with tail index $2\lambda_1$. The asymptotic tail expansion is given by

$$f_W(x) \sim \frac{\Gamma(\lambda_1 + \frac{1}{2})}{\sqrt{2\pi}\,\sigma\,\mathrm{B}(\lambda_1, \lambda_2)} \left[\frac{2\sigma^2}{(x-\mu)^2}\right]^{\lambda_1 + \frac{1}{2}} \quad \text{as} \quad |x| \to \infty.$$

Moment existence conditions follow directly from power-law tail properties: finite variance is guaranteed for $\lambda_1 > 1$, and finite kurtosis is guaranteed for $\lambda_1 > 2$. Simple derivation yields $\mathrm{Var}(W) = \sigma^2\lambda_2/(\lambda_1 - 1)$ and kurtosis $\mathrm{Kurt}(W) = 3(\lambda_1 + \lambda_2 - 1)/[\lambda_2(\lambda_1 - 2)]$. $\square$

S1.2 Gradient and Hessian matrix of $\log$ Ge-N$(x|\boldsymbol{\theta})$

Following the end of Section 3.3, we derive the gradient and Hessian matrix of $\log \text{Ge-N}(x|\boldsymbol{\theta})$, where $\boldsymbol{\theta} = \{\mu, \sigma^2, \boldsymbol{\lambda}\}$. For simplicity, we define $E(\tau) \triangleq E_{\tau|x,\boldsymbol{\theta}}(\tau)$,

$$\mathbf{s}_\tau = \mathbf{s}(\tau|\boldsymbol{\lambda}) \triangleq \underbrace{\nabla_{\boldsymbol{\lambda}} \log f_\tau(\tau|\boldsymbol{\lambda})}_{m\times 1} \quad \text{and} \quad \mathbf{H}_\tau = \mathbf{H}(\tau|\boldsymbol{\lambda}) \triangleq \underbrace{\nabla^2_{\boldsymbol{\lambda}} \log f_\tau(\tau|\boldsymbol{\lambda})}_{m\times m}.$$

The gradient and Hessian matrix of $\log \text{Ge-N}(x|\boldsymbol{\theta})$ are

$$\nabla \log \text{Ge-N}(x|\boldsymbol{\theta}) = \begin{pmatrix} \dfrac{x-\mu}{\sigma^2} E(\tau) \\ -\dfrac{1}{2\sigma^2} + \dfrac{(x-\mu)^2}{2\sigma^4} E(\tau) \\ E(\mathbf{s}_\tau) \end{pmatrix}$$

and

$$\nabla^2 \log \text{Ge-N}(x|\boldsymbol{\theta}) = \begin{pmatrix} H_{\mu\mu} & H_{\mu\sigma^2} & \mathbf{H}_{\mu\boldsymbol{\lambda}}^\top \\ * & H_{\sigma^2\sigma^2} & \mathbf{H}_{\sigma^2\boldsymbol{\lambda}}^\top \\ * & * & \mathbf{H}_{\boldsymbol{\lambda}\boldsymbol{\lambda}} \end{pmatrix},$$

respectively, where

$$\begin{aligned}
H_{\mu\mu} &= -\frac{E(\tau)}{\sigma^2} + \frac{(x-\mu)^2}{\sigma^4}\,\mathrm{Var}(\tau),\\
H_{\mu\sigma^2} &= -\frac{x-\mu}{\sigma^4}\,E(\tau) + \frac{(x-\mu)^3}{2\sigma^6}\,\mathrm{Var}(\tau), \quad \mathbf{H}_{\mu\boldsymbol{\lambda}}^{\top} = \frac{x-\mu}{\sigma^2}\,\mathrm{Cov}(\tau, \mathbf{s}_\tau^{\top}),\\
H_{\sigma^2\sigma^2} &= \frac{1}{2\sigma^4} - \frac{(x-\mu)^2}{\sigma^6}\,E(\tau) + \frac{(x-\mu)^4}{4\sigma^8}\,\mathrm{Var}(\tau),\\
\mathbf{H}_{\sigma^2\boldsymbol{\lambda}}^{\top} &= \frac{(x-\mu)^2}{2\sigma^4}\,\mathrm{Cov}(\tau, \mathbf{s}_\tau^{\top}), \quad \mathbf{H}_{\boldsymbol{\lambda}\boldsymbol{\lambda}} = E(\mathbf{H}_\tau) + \mathrm{Var}(\mathbf{s}_\tau).
\end{aligned}$$

All variances and covariances are computed under the conditional distribution $E(\cdot)$, i.e., $\mathrm{Var}(\tau) = E(\tau^2) - [E(\tau)]^2$, $\mathrm{Cov}(\tau, \mathbf{s}_\tau^{\top}) = E(\tau \mathbf{s}_\tau^{\top}) - E(\tau)E(\mathbf{s}_\tau)^{\top}$, and $\mathrm{Var}(\mathbf{s}_\tau) = E(\mathbf{s}_\tau \mathbf{s}_\tau^{\top}) - E(\mathbf{s}_\tau)E(\mathbf{s}_\tau)^{\top}$.

S2. Statistical inference methods for three specific scale mixtures of normal distribution

S2.1 The inverse gamma mixture of normal distribution

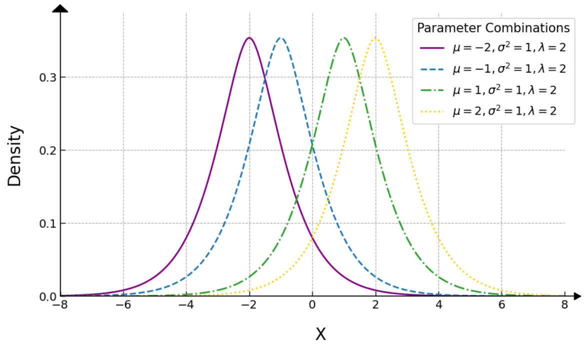
S2.1.1 Basic properties of the IGa-N distribution

Let $f_\tau(\tau|\boldsymbol{\lambda}) = \mathrm{IGamma}(\tau|\lambda, 1)$ in (3.1), we obtain the pdf of $X \sim \text{IGa-N}(\mu, \sigma^2, \lambda)$ as

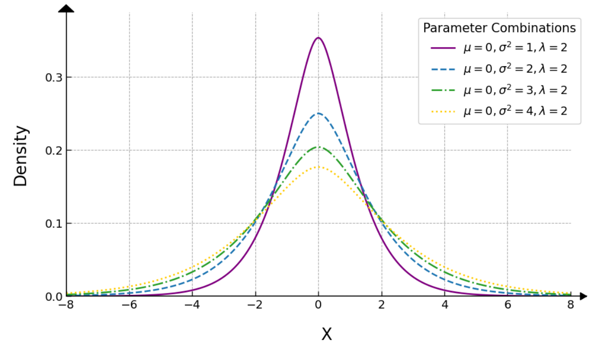
$$\begin{aligned}
\text{IGa-N}(x|\mu, \sigma^2, \lambda) &= \frac{1}{\sqrt{2\pi}\sigma\Gamma(\lambda)} \int_0^\infty \tau^{-\lambda-\frac{1}{2}} \exp\left[-\frac{1}{\tau} - \frac{\tau(x-\mu)^2}{2\sigma^2}\right] \mathrm{d}\tau\\
&\overset{(\text{S5.2})}{=} \frac{1}{\sqrt{2\pi}\sigma\Gamma(\lambda)} [C_0(\alpha, 2, -\lambda+1/2)]^{-1}\\
&\overset{(\text{S5.3})}{=} \frac{2^{-\lambda/2+3/4}}{\sqrt{\pi}\sigma^{\lambda+1/2}\Gamma(\lambda)} |x-\mu|^{\lambda-1/2} K_{-\lambda+1/2}(\sqrt{2}|x-\mu|/\sigma),
\end{aligned}$$

which is symmetric about $x = \mu$, where $C_0(\alpha, 2, -\lambda + 1/2)$ is defined by (S5.2) or (S5.3), $\alpha \triangleq (x-\mu)^2/\sigma^2$, $K_\gamma(\cdot)$ is the modified Bessel function of the second kind defined by (S5.4).

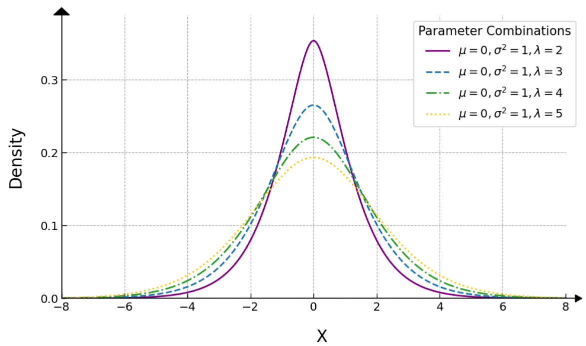
Since $\tau \sim \mathrm{IGamma}(\lambda, 1)$, by direct calculation we have $E(\tau^{-1}) = \lambda$ and $E(\tau^{-2}) = \lambda^2 + \lambda$, so that $E(X) = \mu$, $\mathrm{Var}(X) \overset{(3.3)}{=} \sigma^2\lambda$ and $\mathrm{Kurt}(X) \overset{(3.3)}{=} 3/\lambda$. We found that $\mathrm{Var}(X)$ is an increasing function of σ^2 for a fixed λ, and is also an increasing function of λ for a fixed σ^2.



(a) $\mu = -2, -1, 1, 2, \sigma^2 = 1$ and $\lambda = 2$



(b) $\mu = 0, \sigma^2 = 1, 2, 3, 4$ and $\lambda = 2$



(c) $\mu = 0, \sigma^2 = 1$ and $\lambda = 2, 3, 4, 5$

Figure S1: The density functions of IGa-N(μ, σ^2, λ) under various parameter combinations.

S2.1.2 MLEs of parameters

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim}$ IGa-N(μ, σ^2, λ) and $Y_{\text{obs}} = \{x_i\}_{i=1}^n$ denote the observations, where x_i is a realization of X_i. The log-likelihood function of $\boldsymbol{\theta} = (\mu, \sigma^2, \lambda)^\top$ is given by

$$\ell_{1*}(\boldsymbol{\theta}|Y_{\text{obs}}) = c_{1*} - \frac{n}{2}\log\sigma^2 - n\log\Gamma(\lambda) + \sum_{i=1}^n \log\left[\int_0^\infty h_{1*}(\tau|x_i, \boldsymbol{\theta})\,\mathrm{d}\tau\right],$$

where c_{1*} is a constant, $h_{1*}(\tau|x_i, \boldsymbol{\theta}) = \tau^{-\lambda-1/2}\exp(-\tau^{-1} - \tau\alpha_i/2)$ for $\tau > 0$, and $\alpha_i \triangleq (x_i - \mu)^2/\sigma^2$. We define

$$g_{1*}(\tau|x_i, \boldsymbol{\theta}) \triangleq \frac{h_{1*}(\tau|x_i, \boldsymbol{\theta})}{\int_0^\infty h_{1*}(t|x_i, \boldsymbol{\theta})\,\mathrm{d}t} = \mathrm{GIGau}(\tau|\alpha_i, 2, -\lambda + 1/2), \quad \tau > 0,$$

where $\mathrm{GIGau}(\tau|\alpha_i, 2, -\lambda+1/2)$ denotes the GIGau density defined by (S5.1), and by adopting the N-EM algorithm (Tian *et al.* 2026 [6]), we obtain

$$Q_{1*}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) = c_{1*}^{(t)} - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \mu)^2 a_{1*,i}^{(t)} - n\log\Gamma(\lambda) - n\bar{b}_{1*}^{(t)}\lambda,$$

where $c_{1*}^{(t)}$ is a constant free from $\boldsymbol{\theta}$,

$$a_{1*,i}^{(t)} \triangleq \int_0^\infty \tau \cdot g_{1*}(\tau|x_i, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau \overset{(\text{S5.5})}{=} \frac{\sqrt{2}K_{-\lambda+3/2}(\sqrt{2\alpha_i})}{\sqrt{\alpha_i}K_{-\lambda+1/2}(\sqrt{2\alpha_i})},\ i = 1, \ldots, n, \quad \text{and} \tag{S2.1}$$

$$\bar{b}_{1*}^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n \int_0^\infty \log\tau \cdot g_{1*}(\tau|x_i, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau. \tag{S2.2}$$

Similar to (3.7), we obtain

$$\mu^{(t+1)} = \frac{\sum_{i=1}^n x_i a_{1*,i}^{(t)}}{\sum_{j=1}^n a_{1*,j}^{(t)}} = \sum_{i=1}^n x_i p_{1*,i}^{(t)} = \boldsymbol{x}^\top \boldsymbol{p}_{1*}^{(t)}, \tag{S2.3}$$

$$\sigma^{2(t+1)} = \frac{\sum_{i=1}^n (x_i - \mu^{(t+1)})^2 a_{1*,i}^{(t)}}{n} = \frac{(\boldsymbol{x} - \mu^{(t+1)}\mathbf{1}_n)^\top \mathbf{A}_{1*}^{(t)}(\boldsymbol{x} - \mu^{(t+1)}\mathbf{1}_n)}{n}, \tag{S2.4}$$

$$\lambda^{(t+1)} = \text{sol}\left\{-\psi(\lambda) - \bar{b}_{1*}^{(t)} = 0,\ \forall\,\lambda > 0\right\} \tag{S2.5}$$

$$\overset{(\text{S4.3})}{=} \frac{q_*^{(t)} + \sqrt{q_*^{(t)2} + 2\pi^2/3}}{\pi^2/3}, \tag{S2.6}$$

where $a_{1*,i}^{(t)}$ is given by (S2.1), $p_{1*,i}^{(t)} = a_{1*,i}^{(t)}/\sum_{j=1}^n a_{1*,j}^{(t)}$ for $i = 1, \ldots, n$, $\boldsymbol{p}_{1*}^{(t)} \triangleq (p_{1*,1}^{(t)}, \ldots, p_{1*,n}^{(t)})^\top$, $\boldsymbol{x} = (x_1, \ldots, x_n)^\top$, $\mathbf{A}_{1*}^{(t)} = \text{diag}(a_{1*,1}^{(t)}, \ldots, a_{1*,n}^{(t)})$, $\psi(\lambda) \triangleq \Gamma'(\lambda)/\Gamma(\lambda)$ denotes the digamma function, $\bar{b}_{1*}^{(t)}$ is defined by (S2.2), and $q_*^{(t)} \overset{(\text{S4.4})}{=} -\bar{b}_{1*}^{(t)} - \psi(\lambda^{(t)}) + \pi^2\lambda^{(t)}/6 - 1/\lambda^{(t)}$ by employing US algorithm (Li *et al.* 2026 [4]).

S2.1.3 The IGa-N mean regression model

Let $\{X_i\}_{i=1}^n \overset{\text{ind}}{\sim} \text{IGa-N}(\mu_i, \sigma^2, \lambda)$ and $Y_{\text{obs}} = \{x_i, \boldsymbol{z}_{(i)}\}_{i=1}^n$ denote the observations, we consider the following IGa-N mean regression model: $E(X_i) = \mu_i = \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta}$ for $i = 1, \ldots, n$, where $\boldsymbol{z}_{(i)} = (z_{i1}, \ldots, z_{ip})^\top$ is a p-dimensional vector of covariates for the i-th subject, and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^\top$ is the vector of unknown regression coefficients with $p + 2 \leqslant n$. The log-likelihood function of $\boldsymbol{\theta} = \{\boldsymbol{\beta}, \sigma^2, \lambda\}$ is

$$\ell_1(\boldsymbol{\theta}|Y_{\text{obs}}) = c_1 - \frac{n}{2}\log\sigma^2 - n\log\Gamma(\lambda) + \sum_{i=1}^n \log\left[\int_0^\infty h_1(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta})\,\mathrm{d}\tau\right],$$

where c_1 is a constant, $h_1(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta}) = \tau^{-\lambda-1/2}\exp(-\tau^{-1} - \tau\alpha_i^*/2)$ for $\tau > 0$, and $\alpha_i^* = (x_i - \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta})^2/\sigma^2$. By defining

$$g_1(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta}) \triangleq \frac{h_1(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta})}{\int_0^\infty h_1(t|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta})\,\mathrm{d}t} = \text{GIGau}(\tau|\alpha_i^*, 2, -\lambda + 1/2), \quad \tau > 0,$$

we obtain

$$Q_1(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) = c_1^{(t)} - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n a_{1i}^{(t)}(x_i - \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta})^2 - n\log\Gamma(\lambda) - n\bar{b}_1^{(t)}\lambda,$$

where $c_1^{(t)}$ is a constant free from $\boldsymbol{\theta}$, and

$$a_{1i}^{(t)} \triangleq \int_0^\infty \tau\cdot g_1(\tau|x_i,\boldsymbol{z}_{(i)},\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau \overset{(\text{S5.5})}{=} \frac{\sqrt{2}K_{-\lambda+3/2}(\sqrt{2\alpha_i^*})}{\sqrt{\alpha_i^*}K_{-\lambda+1/2}(\sqrt{2\alpha_i^*})}, \quad i=1,\ldots,n, \quad \text{and} \tag{S2.7}$$

$$\bar{b}_1^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n\int_0^\infty \log\tau\cdot g_1(\tau|x_i,\boldsymbol{z}_{(i)},\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau. \tag{S2.8}$$

Similar to (3.10) and (S2.3)–(S2.6), we obtain

$$\begin{aligned}
\boldsymbol{\beta}^{(t+1)} &= (\mathbf{Z}^\top\mathbf{A}_1^{(t)}\mathbf{Z})^{-1}\sum_{i=1}^n \boldsymbol{z}_{(i)}a_{1i}^{(t)}x_i = (\mathbf{Z}^\top\mathbf{A}_1^{(t)}\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{A}_1^{(t)}\boldsymbol{x},\\
\sigma^{2(t+1)} &= \frac{\sum_{i=1}^n a_{1i}^{(t)}(x_i - \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta}^{(t+1)})^2}{n} = \frac{(\boldsymbol{x}-\mathbf{Z}\boldsymbol{\beta}^{(t+1)})^\top\mathbf{A}_1^{(t)}(\boldsymbol{x}-\mathbf{Z}\boldsymbol{\beta}^{(t+1)})}{n},\\
\lambda^{(t+1)} &= \text{sol}\left\{-\psi(\lambda) - \bar{b}_1^{(t)} = 0,\ \forall\,\lambda>0\right\} \overset{(\text{S4.3})}{=} \frac{q^{(t)} + \sqrt{q^{(t)2} + 2\pi^2/3}}{\pi^2/3},
\end{aligned}$$

where $\mathbf{Z}^\top \triangleq (\boldsymbol{z}_{(1)},\ldots,\boldsymbol{z}_{(n)})$, $\mathbf{A}_1^{(t)} \triangleq \text{diag}(a_{11}^{(t)},\ldots,a_{1n}^{(t)})$, $a_{1i}^{(t)}$ is given by (S2.7), $\boldsymbol{x} = (x_1,\ldots,x_n)^\top$, $\bar{b}_1^{(t)}$ is defined by (S2.8), and $q^{(t)} \overset{(\text{S4.4})}{=} -\bar{b}_1^{(t)} - \psi(\lambda^{(t)}) + \pi^2\lambda^{(t)}/6 - 1/\lambda^{(t)}$.

S2.2 The inverse Gaussian mixture of normal distribution

S2.2.1 Basic properties of the IGau-N distribution

Setting $f_\tau(\tau|\boldsymbol{\lambda}) = \text{IGau}(\tau|\lambda, 1)$ in (3.1), we obtain the pdf of $X \sim \text{IGau-N}(\mu,\sigma^2,\lambda)$ as

$$\begin{aligned}
\text{IGau-N}(x|\mu,\sigma^2,\lambda) &= \frac{1}{2\pi\sigma}\int_0^\infty \tau^{-1}\exp\left[-\frac{\tau(x-\mu)^2}{2\sigma^2} - \frac{(\tau-\lambda)^2}{2\lambda^2\tau}\right]\mathrm{d}\tau\\
&\overset{(\text{S5.2})}{=} \frac{\exp(\lambda^{-1})}{2\pi\sigma}[C_0(\alpha,1,0)]^{-1}\\
&\overset{(\text{S5.3})}{=} \frac{\exp(\lambda^{-1})}{\pi\sigma}K_0\Big(\sqrt{(x-\mu)^2/\sigma^2 + 1/\lambda^2}\Big),
\end{aligned}$$

which is symmetric about $x = \mu$, where $C_0(\alpha,1,0)$ is defined by (S5.2) or (S5.3), $\alpha \triangleq (x-\mu)^2/\sigma^2 + 1/\lambda^2$, $K_0(\cdot)$ is a special modified Bessel function of the second kind defined by (S5.4).

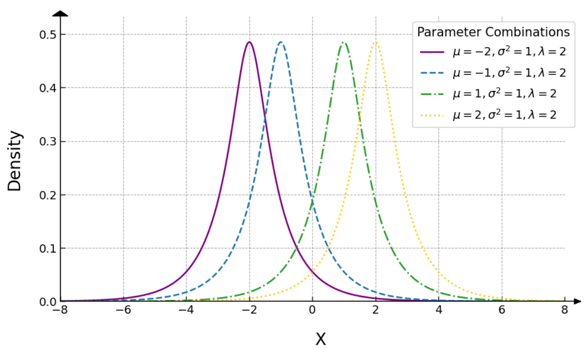
Since the mgf of the $\tau \sim \text{IGau}(\lambda, 1)$ is

$$M_\tau(t) = \exp\left(\frac{1 - \sqrt{1 - 2\lambda^2 t}}{\lambda}\right) \quad \text{and} \quad E(\tau^{-1}) = \frac{\lambda + 1}{\lambda},$$

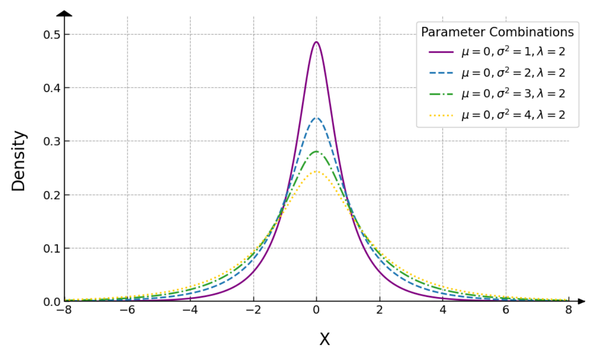
from (3.3) we obtain

$$E(X) = \mu, \;\; \text{Var}(X) = \frac{(\lambda+1)\sigma^2}{\lambda} \quad \text{and} \quad \text{Kurt}(X) = \frac{6\lambda^2 + 3\lambda}{(\lambda+1)^2}. \tag{S2.9}$$

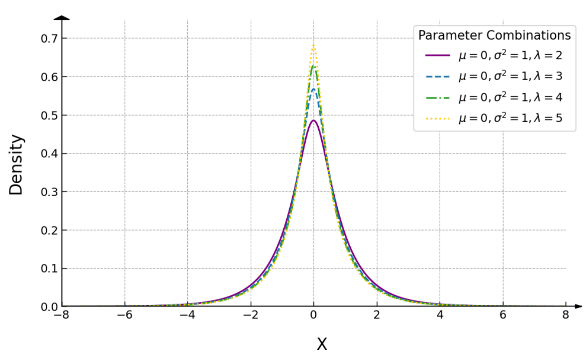
From (S2.9), we can see that $\text{Var}(X)$ is an increasing function of σ^2 for a fixed λ, while a decreasing function of λ for a given σ^2.



(a) $\mu = -2, -1, 1, 2, \sigma^2 = 1$ and $\lambda = 2$



(b) $\mu = 0, \sigma^2 = 1, 2, 3, 4$ and $\lambda = 2$



(c) $\mu = 0, \sigma^2 = 1$ and $\lambda = 2, 3, 4, 5$

Figure S2: The density functions of $\text{IGau-N}(\mu, \sigma^2, \lambda)$ under various parameter combinations.

S2.2.2 MLEs of parameters

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{IGau-N}(\mu, \sigma^2, \lambda)$ and $Y_{\text{obs}} = \{x_i\}_{i=1}^n$ denote the observations, where x_i is a realization of X_i. The log-likelihood function of $\boldsymbol{\theta} = (\mu, \sigma^2, \lambda)^\top$ is given by

$$\ell_{2*}(\boldsymbol{\theta}|Y_{\text{obs}}) \;\; = \;\; c_{2*} - \frac{n}{2}\log\sigma^2 + \sum_{i=1}^n \log\left[\int_0^\infty h_{2*}(\tau|x_i, \boldsymbol{\theta})\,\mathrm{d}\tau\right],$$

where c_{2*} is a constant, and

$$h_{2*}(\tau|x_i,\boldsymbol{\theta}) \triangleq \tau^{-1}\exp\left[-\frac{\tau(x_i-\mu)^2}{2\sigma^2}-\frac{(\tau-\lambda)^2}{2\lambda^2\tau}\right], \quad \tau>0.$$

By defining

$$g_{2*}(\tau|x_i,\boldsymbol{\theta}) \triangleq \frac{h_{2*}(\tau|x_i,\boldsymbol{\theta})}{\int_0^\infty h_{2*}(t|x_i,\boldsymbol{\theta})\,\mathrm{d}t} = \mathrm{GIGau}(\tau|\alpha_i,1,0), \quad \tau>0,$$

where $\mathrm{GIGau}(\cdot|\alpha_i,1,0)$ denotes the GIGau density defined by (S5.1) with $\alpha_i=(x_i-\mu)^2/\sigma^2+1/\lambda^2$, we can similarly construct

$$Q_{2*}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) = c_{2*}^{(t)} - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n (x_i-\mu)^2 a_{2*,i}^{(t)} + \frac{n}{\lambda} - \frac{n\bar{a}_{2*}^{(t)}}{2\lambda^2},$$

where $c_{2*}^{(t)}$ is a constant free from $\boldsymbol{\theta}$,

$$a_{2*,i}^{(t)} \triangleq \int_0^\infty \tau\cdot g_{2*}(\tau|x_i,\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau \overset{(\mathrm{S5.5})}{=} \frac{K_1(\sqrt{\alpha_i})}{\sqrt{\alpha_i}K_0(\sqrt{\alpha_i})}, \quad i=1,\ldots,n, \quad \text{and} \tag{S2.10}$$

$$\bar{a}_{2*}^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n a_{2*,i}^{(t)}. \tag{S2.11}$$

Similar to (3.7), we obtain

$$\begin{cases} \mu^{(t+1)} = \dfrac{\sum_{i=1}^n x_i a_{2*,i}^{(t)}}{\sum_{j=1}^n a_{2*,j}^{(t)}} = \sum_{i=1}^n x_i p_{2*,i}^{(t)} = \boldsymbol{x}^\top \boldsymbol{p}_{2*}^{(t)}, \\ \sigma^{2(t+1)} = \dfrac{\sum_{i=1}^n (x_i-\mu^{(t+1)})^2 a_{2*,i}^{(t)}}{n} = \dfrac{(\boldsymbol{x}-\mu^{(t+1)}\mathbf{1}_n)^\top \mathbf{A}_{2*}^{(t)}(\boldsymbol{x}-\mu^{(t+1)}\mathbf{1}_n)}{n}, \\ \lambda^{(t+1)} = \bar{a}_{2*}^{(t)}, \end{cases} \tag{S2.12}$$

where $a_{2*,i}^{(t)}$ is defined by (S2.10), $p_{2*,i}^{(t)} = a_{2*,i}^{(t)}/\sum_{j=1}^n a_{2*,j}^{(t)}$ for $i=1,\ldots,n$, $\boldsymbol{p}_{2*}^{(t)}=(p_{2*,1}^{(t)},\ldots,p_{2*,n}^{(t)})^\top$, $\boldsymbol{x}=(x_1,\ldots,x_n)^\top$, $\mathbf{A}_{2*}^{(t)}=\mathrm{diag}(a_{2*,1}^{(t)},\ldots,a_{2*,n}^{(t)})$, and $\bar{a}_{2*}^{(t)}$ is defined by (S2.11).

S2.2.3 The IGau-N mean regression model

Let $\{X_i\}_{i=1}^n \overset{\mathrm{ind}}{\sim} \text{IGau-N}(\mu_i,\sigma^2,\lambda)$ and $Y_{\mathrm{obs}}=\{x_i,\boldsymbol{z}_{(i)}\}_{i=1}^n$ denote the observations, we consider the following IGau-N mean regression model: $E(X_i)=\mu_i=\boldsymbol{z}_{(i)}^\top\boldsymbol{\beta}$ for $i=1,\ldots,n$, where $\boldsymbol{z}_{(i)}=(z_{i1},\ldots,z_{ip})^\top$ is a p-dimensional vector of covariates for the i-th subject, and $\boldsymbol{\beta}=$

$(\beta_1, \ldots, \beta_p)^\top$ is the vector of unknown regression coefficients with $p + 2 \leqslant n$. The log-likelihood function of $\boldsymbol{\theta} \triangleq \{\boldsymbol{\beta}, \sigma^2, \lambda\}$ is

$$\ell_2(\boldsymbol{\theta}|Y_{\text{obs}}) = c_2 - \frac{n}{2}\log\sigma^2 + \sum_{i=1}^{n}\log\left[\int_0^\infty h_2(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta})\,\mathrm{d}\tau\right],$$

where c_2 is a constant and

$$h_2(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta}) \triangleq \tau^{-1}\exp\left[-\frac{\tau(x_i - \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta})^2}{2\sigma^2} - \frac{(\tau-\lambda)^2}{2\lambda^2\tau}\right], \quad \tau > 0.$$

By defining

$$g_2(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta}) \triangleq \frac{h_2(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta})}{\int_0^\infty h_2(t|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta})\,\mathrm{d}t} = \text{GIGau}(\tau|\alpha_i^*, 1, 0), \quad \tau > 0,$$

where $\alpha_i^* = (x_i - \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta})^2/\sigma^2 + 1/\lambda^2$, we can similarly construct the surrogate function as

$$Q_2(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) = c_2^{(t)} - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}a_{2i}^{(t)}(x_i - \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta})^2 + \frac{n}{\lambda} - \frac{n\bar{a}_2^{(t)}}{2\lambda^2},$$

where $c_2^{(t)}$ is a constant free from $\boldsymbol{\theta}$,

$$a_{2i}^{(t)} \triangleq \int_0^\infty \tau\cdot g_2(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau \overset{\text{(S5.5)}}{=} \frac{K_1(\sqrt{\alpha_i^*})}{\sqrt{\alpha_i^*}K_0(\sqrt{\alpha_i^*})}, \quad i = 1, \ldots, n, \quad \text{and} \tag{S2.13}$$

$$\bar{a}_2^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^{n}a_{2i}^{(t)}. \tag{S2.14}$$

Similar to (3.10) and (S2.12), we obtain

$$\begin{cases} \boldsymbol{\beta}^{(t+1)} = (\mathbf{Z}^\top\mathbf{A}_2^{(t)}\mathbf{Z})^{-1}\sum_{i=1}^{n}\boldsymbol{z}_{(i)}a_{2i}^{(t)}x_i = (\mathbf{Z}^\top\mathbf{A}_2^{(t)}\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{A}_2^{(t)}\boldsymbol{x}, \\ \sigma^{2(t+1)} = \dfrac{\sum_{i=1}^{n}a_{2i}^{(t)}(x_i - \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta}^{(t+1)})^2}{n} = \dfrac{(\boldsymbol{x} - \mathbf{Z}\boldsymbol{\beta}^{(t+1)})^\top\mathbf{A}_2^{(t)}(\boldsymbol{x} - \mathbf{Z}\boldsymbol{\beta}^{(t+1)})}{n}, \\ \lambda^{(t+1)} = \bar{a}_2^{(t)}, \end{cases}$$

where $\mathbf{Z}^\top \triangleq (\boldsymbol{z}_{(1)}, \ldots, \boldsymbol{z}_{(n)})$, $\mathbf{A}_2^{(t)} \triangleq \text{diag}(a_{21}^{(t)}, \ldots, a_{2n}^{(t)})$, $a_{2i}^{(t)}$ is given by (S2.13), $\boldsymbol{x} = (x_1, \ldots, x_n)^\top$, and $\bar{a}_2^{(t)}$ is defined by (S2.14).

S2.3 The inverted beta mixture of normal distribution

S2.3.1 Basic properties of the IBeta-N distribution

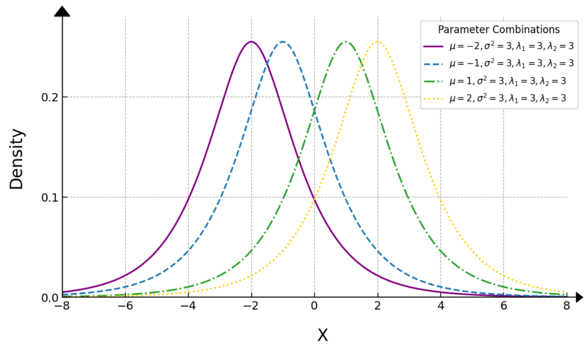
Setting $f_\tau(\tau|\boldsymbol{\lambda}) = \text{IBeta}(\tau|\lambda_1, \lambda_2)$ with $\lambda_1 > 1$ and $\lambda_2 > 1$ in (3.1), we obtain the density function of $X \sim \text{IBeta-N}(\mu, \sigma^2, \lambda_1, \lambda_2)$ as

$$\text{IBeta-N}(x|\mu, \sigma^2, \lambda_1, \lambda_2) = \frac{1}{\sqrt{2\pi}\sigma B(\lambda_1, \lambda_2)} \int_0^\infty \frac{\tau^{\lambda_1 - \frac{1}{2}}}{(1+\tau)^{\lambda_1+\lambda_2}} \exp\left[-\frac{\tau(x-\mu)^2}{2\sigma^2}\right] \text{d}\tau,$$

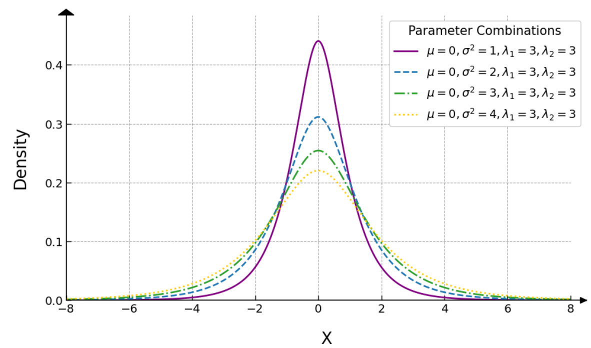
which is symmetric about $x = \mu$. From $\tau \sim \text{IBeta}(\lambda_1, \lambda_2)$ with $\lambda_1 > 1$ and $\lambda_2 > 1$, direct calculation yields $E(\tau^{-1}) = \lambda_2/(\lambda_1 - 1)$ and $E(\tau^{-2}) = (\lambda_2 + 1)\lambda_2/[(\lambda_1 - 1)(\lambda_1 - 2)]$, so that

$$E(X) = \mu,\ \text{Var}(X) \stackrel{(3.3)}{=} \frac{\lambda_2\sigma^2}{\lambda_1 - 1} \quad \text{and} \quad \text{Kurt}(X) \stackrel{(3.3)}{=} \frac{3(\lambda_1 + \lambda_2 - 1)}{(\lambda_1 - 2)\lambda_2}. \tag{S2.15}$$

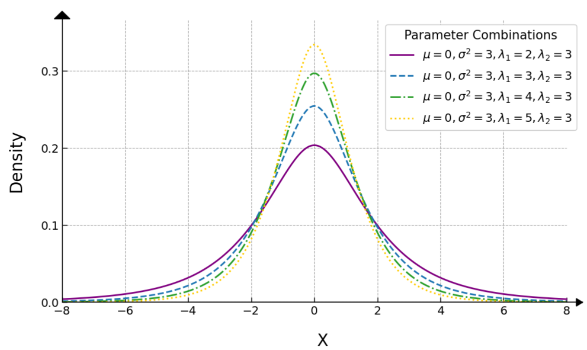
From (S2.15), we can see that $\text{Var}(X)$ is an increasing function of σ^2 for fixed $\{\lambda_1, \lambda_2\}$, also an increasing function of λ_2 for fixed $\{\sigma^2, \lambda_1\}$, while a decreasing function of λ_1 for fixed $\{\sigma^2, \lambda_2\}$.



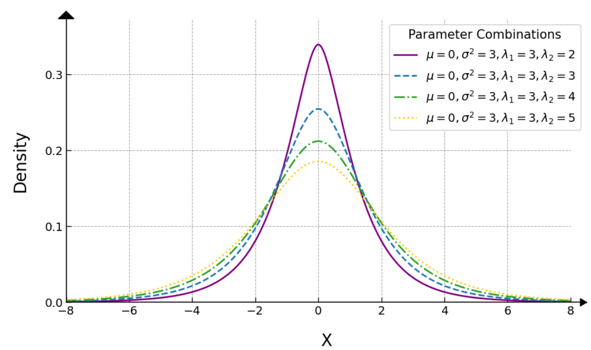
(a) $\mu = -2, -1, 1, 2, \sigma^2 = 3, \lambda_1 = 3$ and $\lambda_2 = 3$



(b) $\mu = 0, \sigma^2 = 1, 2, 3, 4, \lambda_1 = 3$ and $\lambda_2 = 3$



(c) $\mu = 0, \sigma^2 = 3, \lambda_1 = 2, 3, 4, 5$ and $\lambda_2 = 3$



(d) $\mu = 0, \sigma^2 = 3, \lambda_1 = 3$ and $\lambda_2 = 2, 3, 4, 5$

Figure S3: The density functions of IBeta-N$(\mu, \sigma^2, \lambda_1, \lambda_2)$ under various parameter combinations.

S2.3.2 MLEs of parameters

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim}$ IBeta-N$(\mu, \sigma^2, \lambda_1, \lambda_2)$ and $Y_{\text{obs}} = \{x_i\}_{i=1}^n$ be the observations with x_i being the realization of X_i. Then, the log-likelihood function of $\boldsymbol{\theta} \triangleq (\mu, \sigma^2, \lambda_1, \lambda_2)^\top$ is given by

$$\ell_{3*}(\boldsymbol{\theta}|Y_{\text{obs}}) = c_{3*} - \frac{n}{2}\log\sigma^2 - n\log B(\lambda_1, \lambda_2) + \sum_{i=1}^n \log\left[\int_0^\infty h_{3*}(\tau|x_i, \boldsymbol{\theta})\,\mathrm{d}\tau\right],$$

where c_{3*} is a constant, and

$$h_{3*}(\tau|x_i, \boldsymbol{\theta}) \triangleq \frac{\tau^{\lambda_1 - \frac{1}{2}}}{(1+\tau)^{\lambda_1+\lambda_2}} \exp\left[-\frac{\tau(x_i - \mu)^2}{2\sigma^2}\right], \quad \tau > 0.$$

By defining

$$g_{3*}(\tau|x_i, \boldsymbol{\theta}) \triangleq \frac{h_{3*}(\tau|x_i, \boldsymbol{\theta})}{\int_0^\infty h_{3*}(t|x_i, \boldsymbol{\theta})\,\mathrm{d}t}, \quad \tau > 0,$$

we can similarly construct

$$Q_{3*}(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) = c_{3*}^{(t)} - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2 a_{3*,i}^{(t)} - n\log B(\lambda_1,\lambda_2) + (\lambda_1 - 0.5)n\bar{b}_{3*}^{(t)} - (\lambda_1+\lambda_2)n\bar{d}_{3*}^{(t)},$$

where $c_{3*}^{(t)}$ is a constant free from $\boldsymbol{\theta}$,

$$a_{3*,i}^{(t)} \triangleq \int_0^\infty \tau \cdot g_{3*}(\tau|x_i,\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \quad i=1,\ldots,n, \tag{S2.16}$$

$$\bar{b}_{3*}^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^{n}\int_0^\infty \log\tau \cdot g_{3*}(\tau|x_i,\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \quad \text{and} \tag{S2.17}$$

$$\bar{d}_{3*}^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^{n}\int_0^\infty \log(1+\tau) \cdot g_{3*}(\tau|x_i,\boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau. \tag{S2.18}$$

From (3.7) we obtain

$$\mu^{(t+1)} = \frac{\sum_{i=1}^{n} x_i a_{3*,i}^{(t)}}{\sum_{j=1}^{n} a_{3*,j}^{(t)}} = \sum_{i=1}^{n} x_i p_{3*,i}^{(t)} = \boldsymbol{x}^{\top}\boldsymbol{p}_{3*}^{(t)},$$

$$\sigma^{2(t+1)} = \frac{\sum_{i=1}^{n}(x_i-\mu^{(t+1)})^2 a_{3*,i}^{(t)}}{n} = \frac{(\boldsymbol{x}-\mu^{(t+1)}\mathbf{1}_n)^{\top}\mathbf{A}_{3*}^{(t)}(\boldsymbol{x}-\mu^{(t+1)}\mathbf{1}_n)}{n},$$

$$\lambda_1^{(t+1)} = \mathrm{sol}\left\{\psi(\lambda_1+\lambda_2^{(t)}) - \psi(\lambda_1) + \bar{b}_{3*}^{(t)} - \bar{d}_{3*}^{(t)} = 0,\ \forall\ \lambda_1>0\right\} \tag{S2.19}$$

$$\overset{(\text{S4.7})}{=} \frac{q_{1*}^{(t)} + \sqrt{q_{1*}^{(t)2} + 2\pi^2/3}}{\pi^2/3},$$

$$\lambda_2^{(t+1)} = \mathrm{sol}\left\{\psi(\lambda_1^{(t+1)}+\lambda_2) - \psi(\lambda_2) - \bar{d}_{3*}^{(t)} = 0,\ \forall\ \lambda_2>0\right\} \tag{S2.20}$$

$$\overset{(\text{S4.9})}{=} \frac{q_{2*}^{(t)} + \sqrt{q_{2*}^{(t)2} + 2\pi^2/3}}{\pi^2/3},$$

where $a_{3*,i}^{(t)}$ is defined by (S2.16), $p_{3*,i}^{(t)} = a_{3*,i}^{(t)}/\sum_{j=1}^{n} a_{3*,j}^{(t)}$ for $i=1,\ldots,n$, $\boldsymbol{p}_{3*}^{(t)} = (p_{3*,1}^{(t)},\ldots,p_{3*,n}^{(t)})^{\top}$, $\boldsymbol{x} = (x_1,\ldots,x_n)^{\top}$, $\mathbf{A}_{3*}^{(t)} = \mathrm{diag}(a_{3*,1}^{(t)},\ldots,a_{3*,n}^{(t)})$, $q_{1*}^{(t)}$ is given by (S4.8) and $q_{2*}^{(t)}$ is given by (S4.10).

S2.3.3 The IBeta-N mean regression model

Let $\{X_i\}_{i=1}^n \overset{\text{ind}}{\sim}$ IBeta-N$(\mu_i, \sigma^2, \lambda_1, \lambda_2)$ and $Y_{\text{obs}} = \{x_i, \boldsymbol{z}_{(i)}\}_{i=1}^n$ denote the observations, we consider the following IBeta-N mean regression model: $E(X_i) = \mu_i = \boldsymbol{z}_{(i)}^\top \boldsymbol{\beta}$ for $i = 1, \ldots, n$, where $\boldsymbol{z}_{(i)} = (z_{i1}, \ldots, z_{ip})^\top$ is a p-dimensional vector of covariates for the i-th subject and $\boldsymbol{\beta} = (\beta_1, \ldots, \beta_p)^\top$ is the vector of unknown regression coefficients with $p + 3 \leqslant n$. The log-likelihood function of $\boldsymbol{\theta} = \{\boldsymbol{\beta}, \sigma^2, \lambda_1, \lambda_2\}$ is

$$\ell_3(\boldsymbol{\theta}|Y_{\text{obs}}) = c_3 - \frac{n}{2}\log\sigma^2 - n\log B(\lambda_1, \lambda_2) + \sum_{i=1}^n \log\left[\int_0^\infty h_3(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta})\,\mathrm{d}\tau\right],$$

where c_3 is a constant and

$$h_3(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta}) \triangleq \frac{\tau^{\lambda_1 - 0.5}}{(1+\tau)^{\lambda_1+\lambda_2}} \exp\left[-\frac{\tau(x_i - \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta})^2}{2\sigma^2}\right], \quad \tau > 0.$$

By defining

$$g_3(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta}) \triangleq \frac{h_3(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta})}{\int_0^\infty h_3(t|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta})\,\mathrm{d}t}, \quad \tau > 0,$$

we can similarly construct the Q-function as

$$\begin{aligned} Q_3(\boldsymbol{\theta}|\boldsymbol{\theta}^{(t)}) &= c_3^{(t)} - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^n a_{3i}^{(t)}(x_i - \boldsymbol{z}_{(i)}^\top\boldsymbol{\beta})^2 \\ &\quad - n\log B(\lambda_1, \lambda_2) + (\lambda_1 - 0.5)n\bar{b}_3^{(t)} - (\lambda_1 + \lambda_2)n\bar{d}_3^{(t)}, \end{aligned}$$

where $c_3^{(t)}$ is a constant free from $\boldsymbol{\theta}$,

$$a_{3i}^{(t)} \triangleq \int_0^\infty \tau \cdot g_3(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \quad i = 1, \ldots, n, \tag{S2.21}$$

$$\bar{b}_3^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n \int_0^\infty \log\tau \cdot g_3(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau, \quad \text{and} \tag{S2.22}$$

$$\bar{d}_3^{(t)} \triangleq \frac{1}{n}\sum_{i=1}^n \int_0^\infty \log(1+\tau) \cdot g_3(\tau|x_i, \boldsymbol{z}_{(i)}, \boldsymbol{\theta}^{(t)})\,\mathrm{d}\tau. \tag{S2.23}$$

From (3.10) and (S2.19), we obtain

$$
\begin{aligned}
\boldsymbol{\beta}^{(t+1)} &= (\mathbf{Z}^\top \mathbf{A}_3^{(t)} \mathbf{Z})^{-1} \sum_{i=1}^{n} \boldsymbol{z}_{(i)} a_{3i}^{(t)} x_i = (\mathbf{Z}^\top \mathbf{A}_3^{(t)} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{A}_3^{(t)} \boldsymbol{x}, \\
\sigma^{2(t+1)} &= \frac{\sum_{i=1}^{n} a_{3i}^{(t)} (x_i - \boldsymbol{z}_{(i)}^\top \boldsymbol{\beta}^{(t+1)})^2}{n} = \frac{(\boldsymbol{x} - \mathbf{Z}\boldsymbol{\beta}^{(t+1)})^\top \mathbf{A}_3^{(t)} (\boldsymbol{x} - \mathbf{Z}\boldsymbol{\beta}^{(t+1)})}{n}, \\
\lambda_1^{(t+1)} &= \mathrm{sol}\left\{ \psi(\lambda_1 + \lambda_2^{(t)}) - \psi(\lambda_1) + \bar{b}_3^{(t)} - \bar{d}_3^{(t)} = 0 \right\} \overset{\text{(S4.7)}}{=} \frac{q_1^{(t)} + \sqrt{q_1^{(t)2} + 2\pi^2/3}}{\pi^2/3}, \\
\lambda_2^{(t+1)} &= \mathrm{sol}\left\{ \psi(\lambda_1^{(t+1)} + \lambda_2) - \psi(\lambda_2) - \bar{d}_3^{(t)} = 0 \right\} \overset{\text{(S4.9)}}{=} \frac{q_2^{(t)} + \sqrt{q_2^{(t)2} + 2\pi^2/3}}{\pi^2/3},
\end{aligned}
$$

where $\mathbf{Z}^\top \triangleq (\boldsymbol{z}_{(1)}, \ldots, \boldsymbol{z}_{(n)})$, $\mathbf{A}_3^{(t)} \triangleq \mathrm{diag}(a_{31}^{(t)}, \ldots, a_{3n}^{(t)})$, $a_{3i}^{(t)}$ is given by (S2.21), $\boldsymbol{x} = (x_1, \ldots, x_n)^\top$,

$$
\begin{aligned}
q_1^{(t)} &\overset{\text{(S4.8)}}{=} g_3(\lambda_1^{(t)}) + \frac{\pi^2 \lambda_1^{(t)}}{6} - \frac{1}{\lambda_1^{(t)}}, \quad g_3(\lambda_1) = \psi(\lambda_1 + \lambda_2^{(t)}) - \psi(\lambda_1) + \bar{b}_3^{(t)} - \bar{d}_3^{(t)}, \\
q_2^{(t)} &\overset{\text{(S4.10)}}{=} g_4(\lambda_2^{(t)}) + \frac{\pi^2 \lambda_2^{(t)}}{6} - \frac{1}{\lambda_2^{(t)}}, \quad g_4(\lambda_2) = \psi(\lambda_1^{(t+1)} + \lambda_2) - \psi(\lambda_2) - \bar{d}_3^{(t)},
\end{aligned}
$$

$\bar{b}_3^{(t)}$ and $\bar{d}_3^{(t)}$ are defined by (S2.22) and (S2.23), respectively.

S3. Real data analysis

S3.1 Liver disease data

The liver disease dataset was procured from the Department of Hepatobiliary Surgery at a Chinese hospital, encompassing medical records of 1145 patients. Medical research has demonstrated that liver disease patients may exhibit concomitant renal dysfunction, leading to impaired excretion of chloride ions and resultant hyperchloremia. In light of these findings, *Serum Chloride Ion Concentration* (SCIC) in liver disease patients was identified as a pivotal research indicator.

S3.1.1 Parameter estimation

Figure S4 presents the histogram of the SCIC from the liver disease data set, superimposed with graphs of the pdfs derived from the IGa-N, IGau-N, IBeta-N, t, normal and logistic distributions.

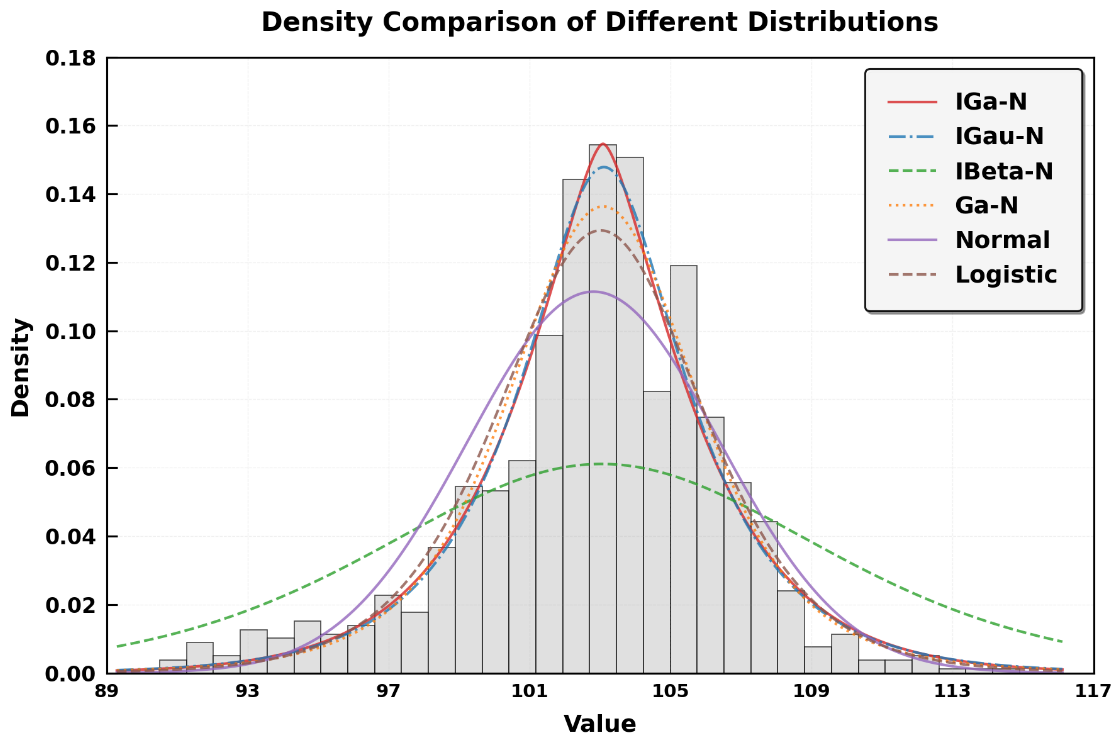


Figure S4: Histogram of the SCIC from the liver disease dataset and the corresponding density curves fitted by the IGa-N, IGau-N, IBeta-N, t, normal and logistic distributions.

Moreover, Table S1 also reports the MLEs of parameters, Std, and two 95% bootstrap CIs for the parameters in six distributions for fitting the SCIC from the liver disease data set.

[Insert Table S1 here]

S3.1.2 Model comparison and selection

Table S2 indicates that while the mean estimates generated by all models are in close proximity to the sample mean of 102.7748, the t distribution demonstrates the highest degree of congruence with the sample mean. The IGau-N model has a variance estimate of 16.2662, closest to the sample variance of 16.3619, indicating it best captures data dispersion. Moreover, IGau-N has the lowest AIC (5765.0970) and BIC (5779.9722), indicating the best goodness-of-fit and complexity balance. Besides, to test the null hypothesis H_0: $\mu_0 = \mu_1$

against H_1: $\mu_0 \neq \mu_1$, where μ_0 represents the sample mean and μ_1 represents the fitted mean, we find all models have p-values above 0.05, so the null hypothesis cannot be rejected that is the fitted mean has statistical significance. In general, the IGau-N model shows the best parameter estimation robustness, complexity control, and fitting performance. It is recommended as the preferred model to analysis the liver disease dataset.

[Insert Table S2 here]

S3.1.3 Mean regression models

In the pursuit of a more comprehensive understanding of the intricate relationship between liver disease and various potential contributing factors, it is imperative to delve deeper into the underlying mechanisms and interactions. Extant research has posited that leukocytosis may exert a detrimental effect on hepatic function, potentially precipitating inflammation and subsequent liver damage. Furthermore, corroborative evidence from pertinent studies has emerged, suggesting that an overabundance of calcium ions may exacerbate hepatocellular injury. An elevated concentration of calcium ions within liver cells is postulated to initiate a cascade of pathological reactions, thereby intensifying the damage inflicted upon hepatocytes.

Consequently, in our forthcoming investigation, we shall meticulously select white blood cell count and calcium ion concentration as salient potential influencing factors. These variables will be incorporated into a regression model, with the objective of discerning their respective contributions and interactions in the context of liver disease. Through this rigorous analytical approach, we aspire to elucidate the complex interplay between these factors and liver disease, thereby advancing our knowledge and potentially informing more targeted therapeutic interventions.

Two covariates for mean regression models are considered as:

Z_2: White blood cell count of patient which are between 1.90 and 7.22 ($\times 10^9$/L), and

Z_3: Value of serum calcium ion concentration which are between 2.08 and 5.87 (mg/dL).

The response variable X_i is the value of SCIC for i-th patient. We consider the following six mean regression models:

$$\begin{aligned} X_i &\stackrel{\text{ind}}{\sim} \text{IGa-N/IGau-N}(\mu_i, \sigma^2, \lambda)/\text{IBeta-N}(\mu_i, \sigma^2, \lambda_1, \lambda_2)/\text{Ga-N}(\mu_i, \sigma^2, \lambda) \\ &\qquad \text{or} \quad N(\mu_i, \sigma^2)/\text{Logistic}(\mu_i, \sigma^2), \quad i = 1, \ldots, n, \quad \text{where} \\ \mu_i &= \beta_1 + z_{i2}\beta_2 + z_{i3}\beta_3. \end{aligned}$$

Among the IGa-N, IGau-N, IBeta-N, t, normal and logistic distributions, the corresponding AIC is 3517.3113, 3516.8745, 3520.6438, 3520.9399, 3531.1676, 3525.0738, and the IGau-N distribution has the smallest AIC as shown in Table S3. The MLE, Std, 95% bootstrap CIs and 95% sandwich CIs of each parameter in the IGau-N mean regression model are represented in Table S3.

[Insert Table S3 here]

S4. Some technical derivations

S4.1 The root of the equation: $-\psi(\lambda) + c_0 = 0$ via a US algorithm

From (S2.5), we know that $\lambda^{(t+1)}$ is the root of the equation:

$$0 = g(\lambda) \triangleq -\psi(\lambda) + c_0, \quad \lambda > 0, \tag{S4.1}$$

where $c_0 \triangleq -\bar{b}_{1*}^{(t)}$ is a known constant. Thus, the aim of this appendix is to solve the root of the equation (S4.1) by employing a US algorithm. The digamma function $\psi(\cdot)$ is defined as

$$\psi(\lambda) \triangleq \frac{\Gamma'(\lambda)}{\Gamma(\lambda)} = -\gamma + \sum_{m=0}^{\infty} \left(\frac{1}{m+1} - \frac{1}{m+\lambda} \right),$$

where $\gamma = \lim_{n\to\infty} \left(\sum_{m=1}^{n} m^{-1} - \log n \right) \approx 0.5772$ is the Euler–Mascheroni constant. Since

$$\begin{aligned} g'(\lambda) &= -\psi'(\lambda) = -\sum_{m=0}^{\infty} \frac{1}{(m+\lambda)^2} = -\frac{1}{\lambda^2} - \sum_{m=1}^{\infty} \frac{1}{(m+\lambda)^2} \\ &> -\frac{1}{\lambda^2} - \sum_{m=1}^{\infty} \frac{1}{m^2} = -\frac{1}{\lambda^2} - \frac{\pi^2}{6} \;\triangleq\; b(\lambda), \end{aligned} \tag{S4.2}$$

the US iteration for calculating $\lambda^{(t+1)}$ is

$$\begin{aligned}
\lambda^{(t+1)} &\overset{(A.3)}{=} \text{sol}\left\{g(\lambda^{(t)}) + \int_{\lambda^{(t)}}^{\lambda} b(z)\,\mathrm{d}z = 0,\ \forall\ \lambda, \lambda^{(t)} > 0\right\} \\
&= \text{sol}\left\{g(\lambda^{(t)}) + \frac{1}{\lambda} - \frac{\pi^2}{6}\lambda - \frac{1}{\lambda^{(t)}} + \frac{\pi^2}{6}\lambda^{(t)} = 0,\ \forall\ \lambda, \lambda^{(t)} > 0\right\} \\
&= \text{sol}\left\{(\pi^2/6)\lambda^2 - q_*^{(t)}\lambda - 1 = 0,\ \forall\ \lambda, \lambda^{(t)} > 0\right\} \\
&= \frac{q_*^{(t)} + \sqrt{q_*^{(t)2} + 2\pi^2/3}}{\pi^2/3},
\end{aligned} \tag{S4.3}$$

where

$$q_*^{(t)} = c_0 - \psi(\lambda^{(t)}) + \frac{\pi^2\lambda^{(t)}}{6} - \frac{1}{\lambda^{(t)}} \quad \text{with } c_0 = -\bar{b}_{1*}^{(t)}. \tag{S4.4}$$

S4.2 Calculation of $\lambda_1^{(t+1)}$ in (S2.19)

From (S2.19), we know that $\lambda_1^{(t+1)}$ is the root of the equation:

$$0 = g_1(\lambda_1) \triangleq \psi(\lambda_1 + \lambda_2^{(t)}) - \psi(\lambda_1) + \bar{b}_{3*}^{(t)} - \bar{d}_{3*}^{(t)}, \quad \lambda_1 > 0, \tag{S4.5}$$

where $\bar{b}_{3*}^{(t)}$ and $\bar{d}_{3*}^{(t)}$ are defined by (S2.17) and (S2.18), respectively. Thus, the aim of this appendix is to solve the root of the equation (S4.5) by employing a US algorithm. From (S4.2), we have

$$\psi'(\lambda) = \sum_{m=0}^{\infty} \frac{1}{(m+\lambda)^2} > 0, \quad \forall \lambda > 0, \tag{S4.6}$$

so that

$$g_1'(\lambda_1) = \psi'(\lambda_1 + \lambda_2^{(t)}) - \psi'(\lambda_1) \overset{(S4.6)}{>} -\psi'(\lambda_1) \overset{(S4.2)}{>} -\frac{1}{\lambda_1^2} - \frac{\pi^2}{6} \triangleq b(\lambda_1).$$

Hence, the US iteration for calculating $\lambda_1^{(t+1)}$ is

$$\begin{aligned}
\lambda_1^{(t+1)} &\overset{(A.3)}{=} \operatorname{sol}\left\{g_1(\lambda_1^{(t)}) + \int_{\lambda_1^{(t)}}^{\lambda_1} b(z)\,\mathrm{d}z = 0,\ \forall\ \lambda_1, \lambda_1^{(t)} > 0\right\} \\
&= \operatorname{sol}\left\{g_1(\lambda_1^{(t)}) + \frac{1}{\lambda_1} - \frac{\pi^2}{6}\lambda_1 - \frac{1}{\lambda_1^{(t)}} + \frac{\pi^2}{6}\lambda_1^{(t)} = 0,\ \forall\ \lambda_1, \lambda_1^{(t)} > 0\right\} \\
&= \operatorname{sol}\left\{(\pi^2/6)\lambda_1^2 - q_{1*}^{(t)}\lambda_1 - 1 = 0,\ \forall\ \lambda_1, \lambda_1^{(t)} > 0\right\} \\
&= \frac{q_{1*}^{(t)} + \sqrt{q_{1*}^{(t)2} + 2\pi^2/3}}{\pi^2/3},
\end{aligned} \tag{S4.7}$$

where

$$q_{1*}^{(t)} = g_1(\lambda_1^{(t)}) + \frac{\pi^2\lambda_1^{(t)}}{6} - \frac{1}{\lambda_1^{(t)}}. \tag{S4.8}$$

S4.3 Calculation of $\lambda_2^{(t+1)}$ in (S2.20)

From (S2.20), we know that $\lambda_2^{(t+1)}$ is the root of the equation:

$$0 = g_2(\lambda_2) \triangleq \psi(\lambda_1^{(t+1)} + \lambda_2) - \psi(\lambda_2) - \bar{d}_{3*}^{(t)}, \quad \lambda_2 > 0,$$

where $\bar{d}_{3*}^{(t)}$ is defined by (S2.18). We have

$$g_2'(\lambda_2) = \psi'(\lambda_1^{(t+1)} + \lambda_2) - \psi'(\lambda_2) \overset{(S4.6)}{>} -\psi'(\lambda_2) \overset{(S4.2)}{>} -\frac{1}{\lambda_2^2} - \frac{\pi^2}{6} \triangleq b(\lambda_2).$$

Hence, the US iteration for calculating $\lambda_2^{(t+1)}$ is

$$\begin{aligned}
\lambda_2^{(t+1)} &\overset{(A.3)}{=} \operatorname{sol}\left\{g_2(\lambda_2^{(t)}) + \int_{\lambda_2^{(t)}}^{\lambda_2} b(z)\,\mathrm{d}z = 0,\ \forall\ \lambda_2, \lambda_2^{(t)} > 0\right\} \\
&= \operatorname{sol}\left\{g_2(\lambda_2^{(t)}) + \frac{1}{\lambda_2} - \frac{\pi^2}{6}\lambda_2 - \frac{1}{\lambda_2^{(t)}} + \frac{\pi^2}{6}\lambda_2^{(t)} = 0,\ \forall\ \lambda_2, \lambda_2^{(t)} > 0\right\} \\
&= \operatorname{sol}\left\{(\pi^2/6)\lambda_2^2 - q_{2*}^{(t)}\lambda_2 - 1 = 0,\ \forall\ \lambda_2, \lambda_2^{(t)} > 0\right\} \\
&= \frac{q_{2*}^{(t)} + \sqrt{q_{2*}^{(t)2} + 2\pi^2/3}}{\pi^2/3},
\end{aligned} \tag{S4.9}$$

where

$$q_{2*}^{(t)} = g_2(\lambda_2^{(t)}) + \frac{\pi^2 \lambda_2^{(t)}}{6} - \frac{1}{\lambda_2^{(t)}}. \tag{S4.10}$$

S5. Generalized inverse Gaussian distribution

A positive r.v. Y is said to follow the *generalized inverse Gaussian* (GIGau) distribution with parameters $\alpha\,(>0)$, $\beta\,(>0)$ and $\gamma\,(\in \mathbb{R})$, denoted by $Y \sim \mathrm{GIGau}(\alpha, \beta, \gamma)$, if its pdf is (Barndorff-Nielsen & Halgreen 1977 [1], Jorgensen 2012 [3])

$$\mathrm{GIGau}(y|\alpha, \beta, \gamma) = C_0(\alpha, \beta, \gamma) \cdot y^{\gamma-1} \exp\left[-\frac{1}{2}\left(\alpha y + \frac{\beta}{y}\right)\right], \quad y > 0, \tag{S5.1}$$

where the normalizing constant

$$C_0(\alpha, \beta, \gamma) \triangleq \left\{\int_0^\infty y^{\gamma-1} \exp\left[-\frac{1}{2}\left(\alpha y + \frac{\beta}{y}\right)\right] \mathrm{d}y\right\}^{-1} \tag{S5.2}$$

has three different expressions depending on the range of parameters (Erdélyi *et al.* 1953 [2]):

$$\begin{aligned} C_0(\alpha, \beta, \gamma) &= \frac{(\alpha/\beta)^{\gamma/2}}{2K_\gamma(\sqrt{\alpha\beta})}, \quad \text{if } \ \alpha, \beta > 0, \ \gamma \in \mathbb{R}, \\ &= \frac{\alpha^\gamma}{2^\gamma \Gamma(\gamma)}, \quad \text{if } \ \alpha > 0, \ \beta = 0, \ \gamma > 0, \\ &= \frac{2^\gamma}{\beta^\gamma \Gamma(-\gamma)}, \quad \text{if } \ \alpha = 0, \ \beta > 0, \ \gamma < 0, \end{aligned} \tag{S5.3}$$

and

$$K_\gamma(z) = \frac{1}{2}\int_0^\infty t^{\gamma-1} \exp\left[-\frac{z}{2}\left(t + \frac{1}{t}\right)\right] \mathrm{d}t = \frac{1}{2}\int_0^\infty t^{-\gamma-1} \exp\left[-\frac{z}{2}\left(t + \frac{1}{t}\right)\right] \mathrm{d}t \tag{S5.4}$$

is the modified Bessel function of the second kind. By direct calculation, we can easily obtain

$$K_\gamma(z) \overset{(S5.4)}{=} K_{-\gamma}(z) \quad \text{and} \quad K_{\gamma+1}(z) = \frac{2\gamma}{z} K_\gamma(z) + K_{\gamma-1}(z).$$

Particularly, when $\gamma = 1/2$, we have

$$K_\gamma(z) = K_{1/2}(z) = K_{-1/2}(z) = \sqrt{\frac{\pi}{2z}} \mathrm{e}^{-z}.$$

Let $Y \sim \text{GIGau}(\alpha, \beta, \gamma)$, then the r-th moment of Y is

$$E(Y^r) = \frac{K_{\gamma+r}(\sqrt{\alpha\beta})}{K_\gamma(\sqrt{\alpha\beta})} \left(\frac{\beta}{\alpha}\right)^{r/2}, \qquad \alpha, \beta > 0, \ \gamma \in \mathbb{R}, \tag{S5.5}$$

where r is an integer. The skewness and kurtosis of Y are respectively given by

$$\begin{aligned} \text{Skew}(Y) &= \frac{K_\gamma^2(\sqrt{\alpha\beta})K_{\gamma+3}(\sqrt{\alpha\beta}) - 3K_\gamma(\sqrt{\alpha\beta})K_{\gamma+1}(\sqrt{\alpha\beta})K_{\gamma+2}(\sqrt{\alpha\beta}) + 2K_{\gamma+1}^3(\sqrt{\alpha\beta})}{[K_\gamma(\sqrt{\alpha\beta})K_{\gamma+2}(\sqrt{\alpha\beta}) - K_{\gamma+1}^2(\sqrt{\alpha\beta})]^{3/2}}, \\ \text{Kurt}(Y) &= \frac{K_\gamma^3(\sqrt{\alpha\beta})K_{\gamma+4}(\sqrt{\alpha\beta}) - 4K_\gamma^2(\sqrt{\alpha\beta})K_{\gamma+1}(\sqrt{\alpha\beta})K_{\gamma+3}(\sqrt{\alpha\beta})}{[K_\gamma(\sqrt{\alpha\beta})K_{\gamma+2}(\sqrt{\alpha\beta}) - K_{\gamma+1}^2(\sqrt{\alpha\beta})]^2} \\ &\quad + \frac{6K_\gamma(\sqrt{\alpha\beta})K_{\gamma+1}^2(\sqrt{\alpha\beta})K_{\gamma+2}(\sqrt{\alpha\beta}) - 3K_{\gamma+1}^4(\sqrt{\alpha\beta})}{[K_\gamma(\sqrt{\alpha\beta})K_{\gamma+2}(\sqrt{\alpha\beta}) - K_{\gamma+1}^2(\sqrt{\alpha\beta})]^2}. \end{aligned}$$

The mgf and the Laplace transform of Y can be derived as

$$\begin{aligned} M_Y(t) &= \left(\frac{\alpha}{\alpha - 2t}\right)^{\gamma/2} \frac{K_\gamma(\sqrt{\alpha\beta - 2t\beta})}{K_\gamma(\sqrt{\alpha\beta})} \quad \text{and} \\ E(\mathrm{e}^{-sY}) &= \left(\frac{\alpha}{\alpha + 2s}\right)^{\gamma/2} \frac{K_\gamma(\sqrt{\alpha\beta + 2s\beta})}{K_\gamma(\sqrt{\alpha\beta})}. \end{aligned}$$

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Table S1 MLE, Std and two 95% bootstrap CIs of parameters for the SCIC from the liver disease data set

Model	Parameter	MLE	Std	l.bound	u.bound	L.bound	U.bould
IGa-N	μ	103.0276	0.1146	102.8029	103.2523	102.8004	103.2482
	σ	3.6252	0.3090	2.9580	4.1693	2.9301	4.1296
	λ	1.1896	0.1813	0.8884	1.5990	0.9904	1.6759
IGau-N	μ	103.0804	0.1131	102.8585	103.3019	102.8560	103.2992
	σ	3.3554	0.2469	2.8574	3.8252	2.8472	3.8201
	λ	2.2482	0.7272	0.9213	3.7721	1.2471	4.0466
IBeta-N	μ	103.0572	0.1210	102.8198	103.2942	102.8172	103.2917
	σ	8.0503	0.3385	7.4002	8.7272	7.3716	8.6749
	λ_1	6.1543	0.5841	4.7065	6.9961	4.9673	7.1492
	λ_2	1.2798	0.4027	0.4494	2.0280	0.5612	2.1503
Ga-N,	μ	103.0850	0.1168	102.8568	103.3146	102.8545	103.3117
t	σ	2.6666	0.2853	2.1128	3.2312	2.1012	3.2204
	λ	1.5632	0.3327	0.9062	2.2104	1.1128	2.2946
N	μ	102.5748	0.1197	102.3531	102.8224	102.3564	102.8191
	σ	4.0431	0.0834	3.8644	4.1917	3.8636	4.1893
Logistic	μ	102.9115	0.1068	102.7040	103.1227	102.7013	103.1204
	σ	2.0851	0.0514	1.9854	2.1872	1.9829	2.1880

Table S2 Fitting the the SCIC (with sample mean 102.5748 and sample variance 16.3619) from liver disease dataset by six distributions, respectively

Model	Mean	Variance	AIC	BIC	p-value (LRT)
IGa-N	103.0276	15.6338	5769.9737	5784.8489	0.0687
IGau-N	103.0804	16.2662	**5765.0970**	**5779.9722**	0.1843
IBeta-N	103.0572	16.1073	5768.1506	5787.9843	0.0726
Ga-N, t	103.0850	19.7360	5770.1273	5785.0025	0.2186
N	102.5748	16.3463	5928.7371	5938.6544	0.1176
Logistic	102.9115	14.3035	5807.3552	5817.2719	0.2191

Table S3 MLE, Std and 95% bootstrap CIs of parameters in the mean regression model for fitting the SCIC from the liver disease data set

Model	Parameter	MLE	Std	95% bootstrap CI	95% sandwich CI	AIC
IGau-N	β_1	102.7480	0.9521	[100.8819, 104.6141]	[100.3221, 105.1740]	
	β_2	-0.1174	0.1479	[-0.4073, 0.1725]	[-0.4942, 0.2594]	
	β_3	0.2032	0.2559	[-0.2984, 0.7048]	[-0.4488, 0.8552]	**3516.8745**
	σ	2.8814	0.2779	[2.3367 3.4261]	[2.1732, 3.5895]	
	λ	1.4862	0.5161	[0.4746, 2.4978]	[0.1706, 2.8017]	